AXIS^2: TWO-DIMENSIONAL LATENT GENERATION FOR LANGUAGE MODELING

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ABSTRACT

Language generation has explored several decoding paradigms. Autoregressive models generate tokens sequentially, while diffusion and masked-token models generate through iterative refinement. More recent latent and hierarchical approaches further move generation away from direct next-token prediction by operating on alternative representations or semantic scales. Despite these differences, generation is typically organized along a single decoding progression. We introduce AXIS^2, which treats generation progress as two-dimensional rather than collapsing it into a single decoding trajectory. Along the first axis, the model progresses through a sequence of representations, ($\mathbf{q}_1 \to \mathbf{q}_2 \to \cdots \to \mathbf{q}_K$), where later representations contain progressively more information needed to recover the final output. Along the second axis, generation within each representation level ($\mathbf{q}_k$) proceeds over multiple rounds: each round generates a subset of the remaining variables conditioned on those generated in previous rounds. Together, the two axes perform complementary refinement in latent space: one progressively increases the information represented about the output, while the other strengthens dependencies among latent variables at the same level. Their combination yields increasingly detailed and better-coordinated latent representations before they are decoded into the final output.Across four continuation benchmarks, AXIS^2 improves ROUGE-1/2/L over the reported autoregressive, masked-generation, and block-diffusion baselines on Wikipedia, WikiSource, and TinyStories; performance remains mixed on One Billion Words and on distributional similarity. Separate measurements of a 12.8B-token checkpoint give 0.55 seconds per example at batch size one on an H100, using 22 backbone evaluations and 88 lightweight sampler calls per 1,024-token window. Analysis of held-out sequences further shows that prediction difficulty varies across the latent hierarchy and correlates differently with text difficulty at different resolutions.

1 INTRODUCTION

Human language production is not simply a process of making one word-level decision after another. Classical psycholinguistic accounts characterize speech production as a hierarchical and incremental process: speakers can form higher-level intentions about what to communicate before the exact lexical realization is determined, while continuing to specify upcoming content as an utterance unfolds (Levelt, 1989; Lee et al., 2013; Hickok, 2012). For example, when answering a complex question in conversation, a speaker may already have a rough idea of the main point to convey before deciding the exact words used to express it. As the utterance unfolds, upcoming content is further specified while the current content is being realized. Nor is this planning completed before speaking begins; we continue organizing upcoming content while realizing the current one. Thus, although language is ultimately expressed as a linear sequence of words, its production naturally involves decisions at multiple levels of abstraction: higher-level decisions determine *what to say*, while lower-level decisions progressively determine *how to say it*.

Recent work has begun to move beyond standard next-token prediction. On one hand, Multi-Token Prediction (MTP) predicts multiple future tokens at once, increasing parallelism while retaining token-level prediction targets (Gloeckle et al., 2024). On the other hand, diffusion language models relax strict autoregressive ordering by updating many tokens in parallel through iterative denoising

or masked refinement (Li et al., 2022; Sahoo et al., 2024; Nie et al., 2025b). Yet these approaches still operate primarily over variables at the final token resolution: they change how many tokens can be predicted or refined together, but do not explicitly organize generation across multiple levels of representation. This leaves open a complementary question: can language generation itself be structured hierarchically, so that coarse content organization and fine-grained linguistic realization are jointly incorporated into the generative process?

In this work, we introduce **AXIS**^{2}, a two-dimensional coarse-to-fine framework for language generation. Along the first dimension, AXIS^2 generates progressively finer representations instead of directly predicting final tokens, allowing higher-level content structure to be established before increasingly specific semantic and linguistic details are resolved. Along the second dimension, AXIS^2 does not require all variables at a given resolution to be generated simultaneously; it progressively commits subsets of them and conditions subsequent predictions on what has already been resolved, preserving dependencies within each resolution. Together, these two dimensions separate high-level content organization from low-level linguistic realization while retaining local dependency modeling. They also substantially shorten the expensive sequential computation path: a single full-model forward pass can advance many generation variables, reducing repeated model execution and parameter movement. This is particularly valuable for low-batch inference, where repeated full-model evaluations directly contribute to latency and memory-bandwidth cost.

In the reported experiments, AXIS^2 improves ROUGE-1/2/L over the compared baselines on three of four continuation benchmarks, although BD3-LM retains stronger distributional similarity on Wikipedia and WikiSource. Separate H100 measurements of a 12.8B-token checkpoint give 0.55 seconds per example with 22 backbone evaluations and 88 lightweight sampler calls. We report quality and timing experiments separately, and analyze how prediction difficulty is distributed across resolutions.

2 Related Work

Token-Level Language Generation. Most language models generate directly at the output-token level, while differing in how predictions are ordered and parallelized. Autoregressive models generate one token at a time (Bengio et al., 2003; Radford et al., 2019), multi-token prediction predicts several future tokens jointly (Gloeckle et al., 2024), and masked or diffusion-based models iteratively refine multiple token states (Ghazvininejad et al., 2019; Austin et al., 2021; Li et al., 2022; Lou et al., 2024; Sahoo et al., 2024). Recent work scales masked diffusion, combines diffusion with block-wise autoregression, or reduces sampling cost (Nie et al., 2025a;b; Arriola et al., 2025; Sahoo et al., 2025). HDLM further introduces coarse-to-fine semantic states within discrete diffusion (Zhou et al., 2025). These works primarily study how token-level generation can be ordered, parallelized, or iteratively refined, whereas AXIS^2 changes the generative representation itself, generating a hierarchy of latent representations before decoding them into tokens.

Latent Representations in Language Generation. Latent representations may either augment token generation or themselves serve as generative variables. NextLat and HiLP retain token-level generation while introducing latent-prediction objectives (Teoh et al., 2025; Shi et al., 2026), while ConceptLM predicts discrete concepts that guide subsequent token generation (Liu et al., 2026). In contrast, other methods generate latent representations directly. Early work autoregressively generates compressed discrete latent sequences (Kaiser et al., 2018); LCM and CALM autoregress over sentence- or chunk-level continuous representations (LCM team et al., 2024; Shao et al., 2025); and latent diffusion or flow methods generate continuous representations before decoding them into text (Lovelace et al., 2023; Jiang et al., 2026; Hu et al., 2026; Guo et al., 2026; Liang et al., 2026). These works either use latent representations to support token generation or generate latents along sequence positions or diffusion trajectories. AXIS^2 instead makes representation resolution the autoregressive axis, generating progressively more informative representations of the same target sequence.

Discrete Latent Hierarchies and Scale-Wise Generation. Hierarchical computation has been explored through multiscale patches and dynamic chunking (Yu et al., 2023; Pagnoni et al., 2025; Hwang et al., 2026), while discrete autoencoders such as VQ-VAE and residual quantization provide learned latent code spaces (van den Oord et al., 2017; Lee et al., 2022). MaskGIT introduces iterative refinement in discrete representations (Chang et al., 2022), and VAR reformulates image

autoregression as coarse-to-fine next-scale prediction (Tian et al., 2024), with later work incorporating text conditioning and intra-scale refinement (Ma et al., 2024; Kumbong et al., 2025). While these works demonstrate the value of discrete hierarchies and scale-wise generation, they primarily study hierarchical representations, multiscale architectures, or coarse-to-fine generation in vision. AXIS^2 instead uses representation resolution as a generative axis for language and introduces progressive within-resolution commitment as a second axis for dependency modeling.

3 AXIS^2: TWO-DIMENSIONAL LATENT GENERATION

Most language generation methods organize generation along a single progression. Autoregressive models advance over token positions (Radford et al., 2019), while masked-token and discrete diffusion models typically refine variables iteratively at a fixed token resolution (Ghazvininejad et al., 2019; Sahoo et al., 2024). AXIS^2 instead organizes generation over a two-dimensional state space in latent representation space.

3.1 HIERARCHICAL MULTI-RESOLUTION REPRESENTATIONS

Let $\mathbf{x}$ denote a target sequence. A pretrained and frozen tokenizer maps $\mathbf{x}$ into a hierarchy of discrete latent codes $\mathcal{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_K)$, where level k contains L_k latent positions, with $L_1 < \cdots < L_K$, and each position is represented by S product-quantization segments (Jégou et al., 2011).

The levels in $\mathcal{Q}$ should not be interpreted as independent semantic representations. Instead, they provide *incremental information* at progressively finer representation resolutions. We instantiate this hierarchy using residual quantization (Lee et al., 2022; Tian et al., 2024). Let $\mathbf{f}_k$ denote the accumulated latent representation after incorporating levels $1{:}k$. Then $\mathbf{f}_k = \mathbf{f}_{k-1} + U_k(D_k(\mathbf{q}_k))$, where D_k dequantizes the discrete codes at level k, and U_k maps them to the common latent resolution. Thus, $\mathbf{q}_k$ specifies the residual information introduced at resolution k, whereas $\mathbf{f}_k$ represents all information accumulated up to that resolution.

The sequence $\mathbf{f}_1 \to \mathbf{f}_2 \to \cdots \to \mathbf{f}_K$, therefore forms a progressively more informative representation of the target. Early accumulated representations retain only part of the information required to reconstruct $\mathbf{x}$, while later levels add residual information that resolves increasingly fine details. The discrete variables $(\mathbf{q}_1, \dots, \mathbf{q}_K)$ are the quantities generated by AXIS^2 to traverse this hierarchy.

This representation hierarchy provides the first dimension of the AXIS^2 generation space. Rather than committing directly to the final token sequence, the model can progressively move from a partial representation toward one containing sufficient information for final text reconstruction.

3.2 AXIS I: REPRESENTATION RESOLUTION

The first axis advances generation across representation resolutions. Starting from an initial representation $\mathbf{f}_0$, AXIS^2 progressively constructs increasingly informative representations by generating the residual codes at each level, $\mathbf{f}_0 \xrightarrow{\mathbf{q}_1} \mathbf{f}_1 \xrightarrow{\mathbf{q}_2} \cdots \xrightarrow{\mathbf{q}_K} \mathbf{f}_K$. The corresponding latent variables are factorized as

$$p_\theta(\mathcal{Q} \mid \mathbf{c}) = \prod_{k=1}^{K} p_\theta(\mathbf{q}_k \mid \mathbf{q}_{<k}, \mathbf{c}), \tag{1}$$

where $\mathbf{q}_{<k}$ determines the accumulated representation $\mathbf{f}_{k-1}$ defined in Section 3.1.

At resolution k, the model does not regenerate the representation from scratch. Instead, it predicts the residual codes $\mathbf{q}_k$ conditioned on the information already established at coarser resolutions. These codes introduce an additional information increment, $\Delta\mathbf{f}_k = U_k(D_k(\mathbf{q}_k))$,, which updates the accumulated representation as $\mathbf{f}_k = \mathbf{f}_{k-1} + \Delta\mathbf{f}_k$. Progression along Axis I therefore preserves information established at earlier resolutions while successively adding information that was not yet represented. Generation evolves from an under-specified representation toward one containing sufficient detail for final text reconstruction. Later stages operate on an already structured representation and only need to supply the remaining information rather than construct the output from scratch.

This changes the primary progression of generation from traversing output positions to traversing representation resolutions. Crucially, the expensive backbone dependency chain is indexed by

representation resolution rather than directly by output length. Autoregressive decoding requires a sequential full-model dependency chain that grows with the number of output tokens T. AXIS^2, in contrast, advances the backbone along a substantially shorter resolution hierarchy, while many latent variables associated with the same resolution can be processed in parallel. This shifts expensive sequential computation from the token level to the representation-resolution level.

Fully parallel prediction within a representation level, however, would prevent variables in the same $\mathbf{q}_k$ from conditioning on one another. Axis II (Section 3.3) introduces a second generation dimension to recover these within-resolution dependencies without reverting to fully token-level sequential decoding.

3.3 Axis II: Within-Resolution Commitment

Axis I allows many variables at the same representation resolution to be predicted in parallel, but fully parallel prediction prevents variables within that resolution from conditioning on one another. At the other extreme, resolving them one by one would recover a fully sequential generation process and sacrifice much of the efficiency gained from the resolution hierarchy. AXIS^2 therefore introduces a second generation axis that progressively commits variables within each representation level.

Let $\mathcal{I}_k$ denote the set of variables in $\mathbf{q}_k$. During generation, AXIS^2 maintains a sequence of increasingly committed subsets, $\varnothing = \mathcal{M}_{k,0} \subset \mathcal{M}_{k,1} \subset \cdots \subset \mathcal{M}_{k,R_k} = \mathcal{I}_k$, where R_k is the number of within-resolution decoding rounds. The variables newly committed at round r are $\mathcal{A}_{k,r} = \mathcal{M}_{k,r} \setminus \mathcal{M}_{k,r-1}$. Conditioned on the representations established at previous resolutions, generation within resolution k proceeds through a sequence of conditional predictions,

$$\mathbf{q}_{k,\mathcal{A}_{k,r}} \sim p_\phi\big(\cdot \mid \mathbf{h}_k, \mathbf{q}_{k,\mathcal{M}_{k,r-1}}\big)\,, \tag{2}$$

where $\mathbf{h}_k$ denotes the resolution-level context produced by the backbone. The commitment rule determines which predicted variables are added to $\mathcal{M}_{k,r}$ at each round.

This second axis progressively exposes within-resolution dependencies: variables predicted in later rounds can condition on those committed earlier at the same representation resolution. When $R_k = 1$, the entire level is predicted in parallel. Increasing R_k allows more predictions to condition on previously established variables, providing a controllable trade-off between parallelism and dependency modeling without reverting to fully token-level autoregression.

Crucially, within-resolution refinement is decoupled from the expensive resolution-level backbone computation. At resolution k, the backbone first computes $\mathbf{h}_k = F_\theta(\mathbf{q}_{<k}, \mathbf{c})$, which depends only on information established before the current resolution. Because current-resolution codes are not fed back into the backbone, $\mathbf{h}_k$ remains fixed throughout within-resolution generation and can be reused across all commitment rounds.

A lightweight refinement module then updates its predictions conditioned on the cached backbone representation and the current commitment state, $p_\phi\big(\mathbf{q}_{k,\mathcal{A}_{k,r}} \mid \mathbf{h}_k, \mathbf{q}_{k,\mathcal{M}_{k,r-1}}\big)$. Thus, additional commitment rounds introduce only lightweight refinement computation rather than additional full-backbone evaluations. Axis I advances the expensive backbone across representation resolutions, whereas Axis II resolves dependencies within a fixed resolution using the lightweight sampler.

3.4 Learning the Two-Dimensional Generator

Having defined the inference process, we train AXIS^2 on states drawn from the same two-dimensional generation space. A training state is specified by $(k, \mathcal{M}_k)$, where k identifies the current representation resolution along Axis I, and $\mathcal{M}_k \subseteq \mathcal{I}_k$ specifies the variables already committed within that resolution along Axis II.

Given a target sequence $\mathbf{x}$, the frozen tokenizer deterministically produces its representation hierarchy $\mathcal{Q} = (\mathbf{q}_1, \ldots, \mathbf{q}_K)$. For each representation resolution k, the backbone conditions on the representations established at earlier resolutions and produces $\mathbf{h}_k = F_\theta(\mathbf{q}_{<k}, \mathbf{c})$. This trains the resolution-level generator to advance along Axis I: information accumulated at resolutions $< k$ is used to predict the residual information introduced by $\mathbf{q}_k$.

To train Axis II, we sample a valid commitment state $\mathcal{M}_k \subseteq \mathcal{I}_k$ and define the unresolved variables as $\mathcal{U}_k = \mathcal{I}_k \setminus \mathcal{M}_k$. The lightweight refinement module predicts these unresolved variables conditioned

on both the cached backbone representation and the committed variables. For one commitment state, we optimize the normalized prediction loss

$$\mathcal{L}_k = -\frac{1}{|\mathcal{U}_k|} \sum_{i \in \mathcal{U}_k} \log p_\phi(q_{k,i} \mid \mathbf{h}_k, \mathbf{q}_{k,\mathcal{M}_k}) \,. \tag{3}$$

These partially committed states mirror the intermediate states encountered during decoding, training the same refinement module to operate at different points along the within-resolution axis.

To improve supervision efficiency, we reuse each expensive backbone representation across multiple valid commitment states. Specifically, for each backbone forward, we sample $\left\{\mathcal{M}_k^{(m)}\right\}_{m=1}^{M}$, and evaluate the lightweight refinement module on all M states while sharing the same cached $\mathbf{h}_k$. The corresponding resolution-level objective is $\bar{\mathcal{L}}_k = \frac{1}{M}\sum_{m=1}^{M} \mathcal{L}_k^{(m)}$. Here, M controls the number of training commitment states reused per backbone representation; it is independent of the number of refinement rounds R_k used during inference. This increases the amount of supervision obtained from each expensive backbone computation without lengthening the inference trajectory.

Finally, the overall objective aggregates losses across representation resolutions. A flattened objective implicitly gives greater weight to resolutions containing more latent variables, which need not reflect their prediction difficulty. We therefore use a resolution-aware weighting, $\mathcal{L}_{\text{res}} = \sum_{k=1}^{K} \omega_k(\alpha)\, \bar{\mathcal{L}}_k$, where $\omega_k(\alpha)$ controls the relative contribution of different representation resolutions. Experimental settings are summarized in Section 4.1.

4 Experiments

4.1 Experimental Setup

Data and evaluation. We train on OpenWebText2 (OWT2), a component of The Pile (Gao et al., 2020), using GPT-2 BPE (Radford et al., 2019). We evaluate 1,000 recorded prompt–continuation pairs per benchmark on English Wikipedia[1], WikiSource[2], TinyStories (Eldan & Li, 2023), and One Billion Words (1BW) (Chelba et al., 2014). Prompts contain a seeded 40–60% prefix of each source text. AXIS^2 retains at most 1,024 prompt tokens and generates 1,024-token windows, chaining up to four when needed; special tokens are removed on decoding and outputs are truncated to the reference word count. EOS does not stop AXIS^2 generation. The recorded selection/test split cannot be independently reconstructed from the snapshot.

Models and budgets. The main table compares a GPT-2-style AR model (Radford et al., 2019), CMLM with Mask-Predict (Ghazvininejad et al., 2019), and BD3-LM (Arriola et al., 2025). AXIS^2 uses a 12-layer, 768-dimensional, 12-head Transformer (Vaswani et al., 2017): the planner has 87.3M parameters (approximately 85M in the backbone), plus a 4.16M sampler and a separately pretrained, frozen 69.9M tokenizer. Tokenizer pretraining uses approximately 39.3B token positions and is separate from generator training. The main configuration uses $M = 8$ masking states, C2K2 within-resolution refinement, classifier-free guidance (Ho & Salimans, 2022), and scale reweighting with $\alpha = 0.25$. The main results retain the repository's nominal "51.2B" run label; archived steps instead imply about 41.1B target-token exposures for AXIS^2 under the checked-in configuration, preventing a verified equal-token comparison.

Metrics. We report ROUGE-1/2/L (Lin, 2004), BERTScore F1 (Zhang et al., 2020) with RoBERTa-large (Liu et al., 2019), and MAUVE (Pillutla et al., 2021) with GPT-2-large features capped at 256 tokens. Generation perplexity is prompt-conditioned corpus perplexity under GPT-2-large and is reported together with unigram entropy, Distinct-1/2 (Li et al., 2016), and repeated 4-grams. Efficiency measurements distinguish backbone NFE, lightweight sampler calls, single-example time, and batched throughput.

Table 1: Generation quality for the repository's nominal 51.2B setting. The AXIS^2 budget label is not reconciled with archived training steps (Section 4.1), so this is not a verified iso-token comparison. R1/R2/RL and BS are ROUGE-1/2/L and BERTScore F1 ($\times 100$); M is MAUVE@256. Each benchmark contains 1,000 continuations. Best and second-best values among reported methods are **bold** and underlined. Near-floor MAUVE values on TinyStories and 1BW are reported without ranking.

Model	Wikipedia					WikiSource				
	R1	R2	RL	BS	M	R1	R2	RL	BS	M
AR (Radford et al., 2019)	18.63	1.71	10.11	75.42	0.049	22.13	1.92	11.56	76.50	0.031
CMLM (Ghazvininejad et al., 2019)	10.86	1.77	9.37	72.58	0.008	12.64	2.62	10.59	73.65	0.009
BD3-LM (Arriola et al., 2025)	24.09	2.63	12.29	79.21	**0.226**	28.64	3.05	13.99	**80.25**	**0.240**
AXIS^2	**29.06**	**4.82**	**15.00**	**79.36**	0.195	**31.26**	**4.90**	**15.55**	79.91	0.091
Model	TinyStories					1BW				
	R1	R2	RL	BS	M	R1	R2	RL	BS	M
AR (Radford et al., 2019)	15.89	0.88	10.02	77.05	0.006	7.21	0.21	6.29	79.13	0.005
CMLM (Ghazvininejad et al., 2019)	15.62	2.33	11.88	78.06	0.005	9.38	**0.75**	8.95	78.87	0.005
BD3-LM (Arriola et al., 2025)	27.10	2.79	15.18	83.14	0.008	**10.97**	0.62	**9.35**	**82.52**	0.005
AXIS^2	**33.68**	**5.74**	**18.73**	**83.28**	0.007	10.23	0.51	8.75	81.88	0.005

Table 2: Generation diagnostics for the main AXIS^2 checkpoint (1,000 samples per benchmark). PPL is prompt-conditioned GPT-2-large corpus perplexity; entropy is corpus GPT-2 unigram entropy in bits. D1/D2 are corpus distinct-word n-gram ratios. Rep4 is the mean within-sequence repeated-word 4-gram fraction (%), recomputed from the archived samples.

Benchmark	PPL	Entropy	D1	D2	Rep4 (%)
Wikipedia	228.6	11.21	0.229	0.736	0.465
WikiSource	217.9	10.54	0.176	0.675	0.532
TinyStories	102.3	8.81	0.123	0.574	0.238
1BW	626.2	10.40	0.382	0.866	0.043

4.2 Generation Quality

Table 1 shows that AXIS^2 has the highest ROUGE-1/2/L among the reported methods on Wikipedia, WikiSource, and TinyStories. Relative to BD3-LM, its ROUGE-1 gains on these datasets are 4.97, 2.62, and 6.58 points. The comparison is mixed on other metrics: BD3-LM has higher MAUVE on Wikipedia and WikiSource and higher ROUGE-1/L and BERTScore on 1BW, while CMLM has the highest 1BW ROUGE-2. Thus, the overlap gains do not imply uniformly better generation quality.
Generation diagnostics. Table 2 reports Gen-PPL together with entropy, diversity, and repetition for the same AXIS^2 samples used in the quality comparison.

4.3 Inference Efficiency

Table 3 separates expensive backbone evaluations from lightweight refinement calls. On one NVIDIA H100, batch size one takes 0.55 seconds per example; batching 32 examples gives approximately $2.5\times$ higher throughput. These measurements use paired runs of 32 and 96 Wikipedia examples, with time estimated as $(t_{96} - t_{32})/64$ to cancel fixed loading overhead. They concern the token-weighted 12.8B checkpoint, separately from the main quality experiment. The batching path changes random-number ordering, and only one timing pair is available per batch size; no uncertainty interval or matched-baseline speedup is inferred.

[1] https://huggingface.co/datasets/wikimedia/wikipedia
[2] https://huggingface.co/datasets/wikimedia/wikisource

Table 3: Archived H100 timings for the same 12.8B-token AXIS^2 2D C2K2 checkpoint. Calls are per 1,024-token window and count both CFG branches. Time at batch size 32 is amortized time, not single-request latency. Throughput is the inverse of the reported time.

Batch	Backbone NFE	Sampler calls	Time (s/example)	Examples/s
1	22	88	0.55	1.82
32	22	88	0.219	4.57

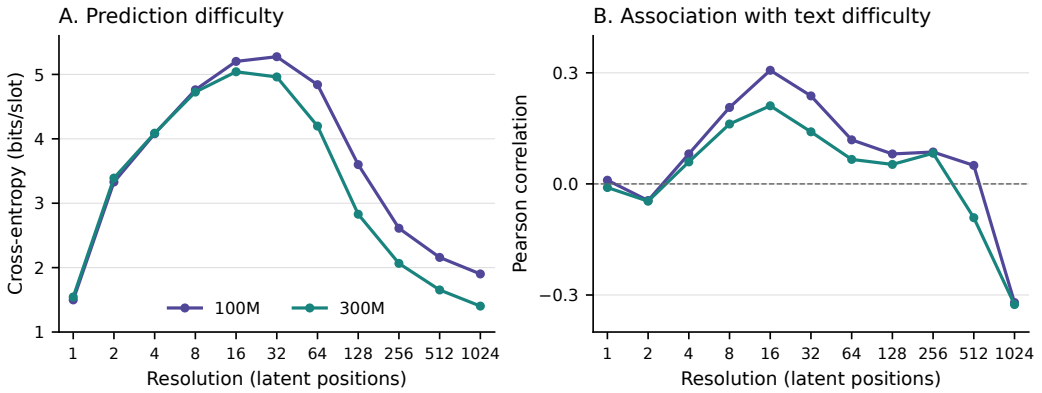


Figure 1: Resolution-wise difficulty on the same 2,048 validation windows. Left: mean prediction cross-entropy; shading denotes a normal-approximation 95% confidence interval for the 100M model. Right: Pearson correlation between resolution-wise cross-entropy and frozen GPT-2-large NLL.

In a separate same-checkpoint decoding sweep, reducing fine-scale commitment from C2K2 to C1K1 lowers sampler calls from 88 to 70 and measured time from 0.54 to 0.50 seconds per example, while Wikipedia ROUGE-1/2 changes from 29.00/4.43 to 28.94/4.37. The backbone cost stays fixed at 22 evaluations, illustrating the distinct computational roles of the two generation axes.

4.4 Content-Adaptive Representation Across Resolutions

We examine 2,048 OpenWebText2 validation windows of 1,024 tokens using the 100M-class model, with a 300M model as a robustness check. The frozen tokenizer provides 11 resolutions, from q_1 to q_{1024}. For each window, we measure teacher-forced cross-entropy at each resolution conditioned on ground-truth coarser codes. An external difficulty measure is the mean token NLL under frozen GPT-2-large (Radford et al., 2019).

Figure 1 shows that prediction difficulty concentrates at intermediate resolutions. For the 100M model, cross-entropy rises from 1.50 bits/slot at q_1 to 5.20 at q_{16} and 5.28 at q_{32}, then falls to 1.90 at q_{1024}. The 300M model retains this overall pattern. The right panel shows that intermediate-resolution losses positively correlate with external text difficulty, peaking at q_{16} ($r = 0.31$), whereas q_{1024} is negatively correlated ($r = -0.32$). The corresponding correlations for 300M are 0.21 and -0.33, respectively. Thus, scale-specific difficulty varies with content even though the resolution grid and decoding schedule remain fixed.

As an illustrative tokenizer diagnostic, progressively decoding ground-truth code prefixes for six windows spanning the external difficulty range yields mean exact-token reconstruction accuracies of 10.4%, 36.1%, 94.8%, and 99.9% through q_{32}, q_{128}, q_{512}, and q_{1024}, respectively. These oracle reconstructions measure information retained by the tokenizer, rather than generation accuracy. Together, the measurements support differentiated prediction problems across resolutions, without implying token-specific computation budgets or a causal semantic–lexical division.

5 CONCLUSION

AXIS^2 combines next-resolution generation with lightweight within-resolution commitment. The reported experiments support a useful trade-off between reference overlap and sequential backbone computation. Distributional quality, matched-workload baseline timing, and complete training-budget and selection records remain limitations of the current evidence.

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